

Speech Privacy Protection & Voice Conversion System

Traditional Signal Processing + Deep Learning Hybrid Architecture

INPUT LAYER

① Single Audio

.wav / .mp3 / .m4a
one audio file
for gender conversion
parametric mode

② Source + Reference

source audio to convert
+ target reference audio
dual-audio input
voice cloning mode

③ Ref + Audio-to-Eval

reference audio
+ audio-to-evaluate
pair comparison
similarity evaluation

④ Source + Voice Pool

source audio (Speaker A)
+ target voice pool
N = 9 candidates
male / female pools
pre-built embeddings
anonymization mode
→ 3 output variants

PROCESSING MODULES

Module 1 — WORLD Single-Audio Voice Conversion

Traditional Signal Processing | Vocoder-Based Parametric Decoupling | source-filter model

DIO+StoneMask

F0 extraction
median → gender
pitch_ratio × F0

CheapTrick

spectral envelope
formant_ratio
frequency warping

D4C

aperiodicity AP
breath / fricatives
preserved as-is

Synthesize

WORLD synthesis
strength blending
 $y = \text{mod} \cdot s + \text{orig} \cdot (1-s)$

Output + Visualization

converted audio (pitch + formant controlled)
F0 histogram: before vs. after
Mel spectrogram comparison

Module 2 — Voice Cloning (Dual-Audio)

Deep Learning | Zero-Shot Timbre Transfer | Speaker Embedding Injection

Path A — FreeVC (Coqui TTS)

PPG ContentEncoder → Speaker Encoder → HiFi-GAN decoder
encoder-decoder | disentangled content & speaker representation
condition: PPG content + target embedding + prosody (F0, duration)
latent space interpolation for stable timbre transfer

Path B — OpenVoice v2

Speaker Embedding extraction → ToneColorConverter
 $\text{style} = (1-\tau) \cdot \text{src_style} + \tau \cdot \text{tgt_style} \quad \tau \in [0.20, 0.40]$
 $\tau \uparrow \rightarrow \text{more like target} \mid \tau \downarrow \rightarrow \text{more natural / stable}$
zero-shot | few-second reference | cross-lingual capable

Module 3 — Speaker Similarity Evaluation

OpenVoice Speaker Encoder | Cosine Similarity | Reused in Anonymization

Extract speaker embeddings from both audios → $\text{cosine}(a, b) = a \cdot b / (\|a\| \|b\|) \rightarrow \text{score mapped to } [0, 100]$

Short audio pad mechanism (<6 s auto-extend to prevent extraction failure)

85–100: highly similar · 70–85: moderately similar · <70: low similarity

Module 4 — Speech Anonymization (Core Module)

PPG Decoupling + Target Pool Selection | VoicePrivacy Framework (ASV-EER / ASR-WER)

Preprocess → ContentEncoder (PPG) + Speaker Embedding (pool) → 3-Method Anonymization → Evaluate

① freevc_baseline

PPG + embedding direct combination
highest de-identification | source_sim=0.125 | WER=0.089

② metric_clarity

score = $\alpha \times \text{SpecDist} + \beta \times \text{EnvCorr}$, select top candidate
clarity-first | best acoustic quality | source_sim=0.146

③ ppg_tone

PPG content fidelity priority + tone adjustment
content-first | envelope closest to original

OUTPUT LAYER

① Converted Audio

pitch + formant controlled
+ F0 histogram
+ Mel spectrogram
before/after comparison
auto gender detect

② Cloned Audio

FreeVC or OpenVoice
converted output
+ Speaker Similarity
score (0 – 100)
+ similarity_*.json
cosine similarity

③ Anon Audios A/B/C

3 anonymized variants
+ VoicePrivacy Report

ASV-EER ↑ [Privacy]

speaker verification EER

ASR-WER ↓ [Utility]

word error rate

source_sim ↓ [Privacy]

cosine sim to source spk

content_kept ↑ [Utility]

1 – WER

pipeline_report.json